10.3.7 Affinity chromatography

Affinity chromatography is an efficient technique, where separation of a biomolecule can be achieved through highly specific interactions. The interaction is based on the chemical nature of the molecule present in the sample.

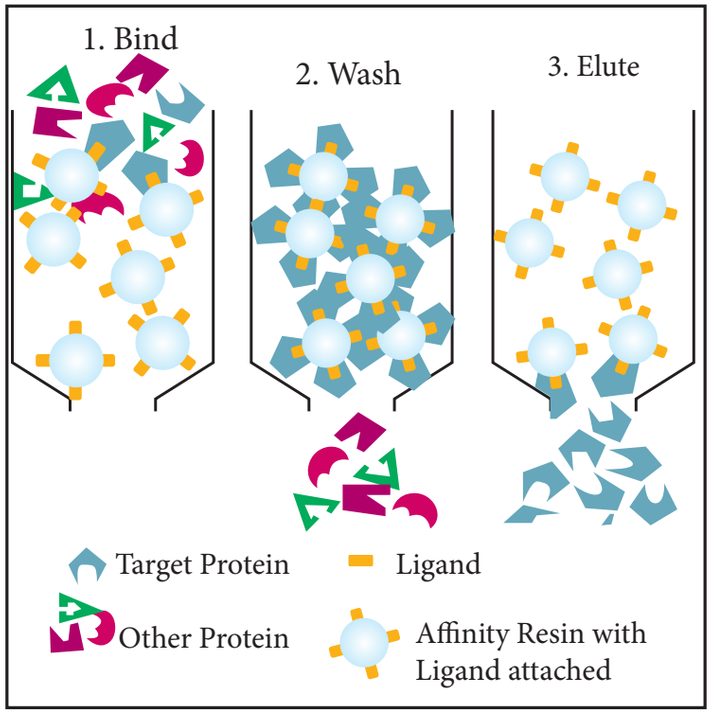


Figure 10.7 Affinity chromatography

When a complex mixture containing the specific compound to be purified is added to the immobilized ligand, only the specific compound will bind reversibly to the ligand leaving the other constituents. The target compound (protein) can be suitably recovered by displacement from the ligand using selective buffer with altered pH (Figure.10.7).

10.3.8 High performance liquid chromatography

HPLC (High Performance Liquid Chromatography) is a highly improved form of liquid column chromatography used to separate the components of a mixture. Columns used in the HPLC system are generally made in stainless steel to withstand the high pressures. Sample administration is through an automated injection system. Mobile phase (mixture of polar or non-polar liquid components) are contained in a glass reservoir. Eluents used for HPLC must be purified. Pumping systems for delivery of the eluent are one of the most important features of HPLC systems. A detector is used to retrieve the data and its analysis in the form of chromatogram (Fig.10.8). Applications of HPLC include isolation and purification of biologically active compounds, purification of chemical compounds, and developing processes for synthesizing chemical compounds.

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